

3.2.3 Activity: Arithmetic Calculator Using Functions

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CS XII

Chapter 3: Programming Fundamentals

Lecture 12 of 18

3.2.3 Activity: Arithmetic Calculator Using Functions

LEARN IT

Each arithmetic operation is a **separate function**. That is modularity: addition is one small task, division is another (with a guard against dividing by zero). The main program only asks for two numbers and calls the functions.

```
def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b != 0:
        return a / b
    else:
        return "Cannot divide by zero"

num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
print("Addition:", add(num1, num2))
print("Subtraction:", subtract(num1, num2))
print("Multiplication:", multiply(num1, num2))
print("Division:", divide(num1, num2))
```

WORKED EXAMPLE

Sample run with 12 and 4. `float` is used so decimals work; even whole numbers print with a decimal (12 becomes 12.0).

```
OUTPUT
Enter first number: 12
Enter second number: 4
Addition: 16.0
Subtraction: 8.0
Multiplication: 48.0
Division: 3.0
```

Check: $12 + 4 = 16$, $12 - 4 = 8$, $12 \times 4 = 48$, $12 \div 4 = 3$. In Python 3, `/` always produces a float, so 3.0 not 3.

Zero divisor:

```
def divide(a, b):
    if b != 0:
        return a / b
    else:
        return "Cannot divide by zero"

print(divide(10, 0))
print(divide(10, 2))
```

```
OUTPUT
Cannot divide by zero
5.0
```

Without the `if b != 0` test, `10 / 0` would raise `ZeroDivisionError` and stop the program.

REMEMBER IT

Four functions, four names: `add`, `subtract`, `multiply`, `divide`. The main program should look like a contents page — a sequence of calls — not a pile of formulae.

COMMON MISTAKE

`input()` returns a string. If you skip `float()` (or `int()`), `add` would concatenate: `"12" + "4"` is `"124"`. Convert first. Also, `divide` returns either a number or a string; do not later do arithmetic on the error message.

اردو میں سمجھیں

کیلکولیٹر میں ہر عمل الگ فنکشن ہے: جمع، تفریق، ضرب، تقسیم۔ یہ ماڈیولریٹی ہے۔ تقسیم سے پہلے صفر کی جانچ ضروری ہے ورنہ پروگرام رک جاتا ہے۔ `input()` سے ملنے والی تار کو `float()` سے عدد بنانا لازمی ہے ورنہ جمع جوڑنے کے بجائے الفاظ جوڑ دے گی۔

CHECK YOURSELF

1. What does `divide(10, 0)` return in this activity?
2. Why are the numbers converted with `float()` ?

Answers: 1. The string "Cannot divide by zero". 2. Because `input()` returns text; `float()` turns it into a number so `+` `-` `*` `/` do arithmetic, including decimals.

Next lecture: 3.2.4 Scope of Local and Global Variables